

Achilles-Bench: A Challenging Benchmark for Low-Resource Evaluation

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Abstract

With promising yet saturated results in high-resource settings, low-resource datasets have gradually become crucial benchmarks (e.g., BigBench Hard, superGLUE) for evaluating the learning ability of advanced neural networks. In this work, we find that there exists a set of “hard examples” in low-resource settings that challenge neural networks but are not well evaluated, which causes over-estimated performance. We first give a theoretical analysis on which factors bring the difficulty of low-resource learning. It then motivates us to propose a challenging benchmark Achilles-Bench to better evaluate the learning ability, which covers 11 datasets, including 8 natural language process (NLP) datasets and 3 computer vision (CV) datasets. Experiments on a wide range of models show that neural networks, even pre-trained language models, have sharp performance drops on our benchmark, demonstrating the effectiveness of evaluating the weaknesses of neural networks. On NLP tasks, we surprisingly find that despite better results on traditional low-resource benchmarks, pre-trained networks, does not show performance improvements on our benchmarks. there is still a large robustness gap between existing models and human-level performance, highlighting the need for robust low-resource learning models.¹

1 Introduction

Large-scale models have shown strong capabilities in learning from a handful of examples (Scao et al., 2022; Touvron et al., 2023a; OpenAI, 2023), resulting in an increased demand for low-resource benchmarks. Numerous research studies have highlighted the rapid adaptability of such models to new tasks, utilizing techniques like in-context learning (Dong et al., 2022). Consequently, the evalua-

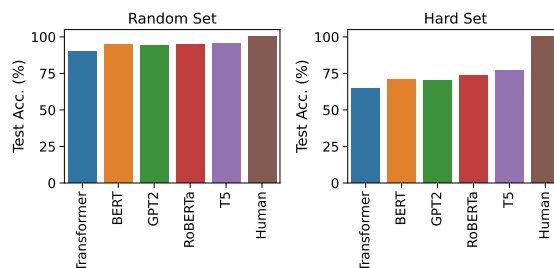


Figure 1: Results on sentiment classification (SST-2). The left figure shows average results on a randomly-sampled set as the test set. The right figure shows average results on a hard set as the test set. The hard test set is selected with smaller loss margins given a weak classifier. Although it is widely-accepted that neural networks can handle sentiment classification well with near-human accuracy (as shown in the left figure), the large drop on hard examples demonstrate that existing models still have generalization issues.

tion of large-scale pre-trained models has shifted towards assessing their ability to quickly learn new downstream tasks with limited available samples, including superGLUE (Wang et al., 2019) and BIG-Bench Hard (Suzgun et al., 2022b).

However, many low-resource datasets usually use random or manual selection methods to sample data from the cleaned and balanced training data. They struggle to capture the data biases and increased difficulty commonly encountered in real-world scenarios. Consequently, these benchmarks fall short in evaluating the true learning gap between existing models and human-level models. While some models can surpass human performance on these benchmarks (e.g., SST-2) (Yang et al., 2019; Nangia and Bowman, 2019; He et al., 2021), many studies have revealed that these robust models still face challenges such as spurious correlation (Sagawa et al., 2020; Hu et al., 2023a) or bias (Bolukbasi et al., 2016), which are relatively uncommon in human learning. As depicted in Figure 1, models on a randomly sampled low-resource set demonstrate performance comparable

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¹Code and data are available on <https://github.com/Qian2333/Achilles-Bench>.

to human-level in sentiment analysis. However, their performance significantly deviates from human level when confronted with challenging examples. It motivates us to propose a challenging low-resource benchmark.

In this work, we aim to find challenging examples given any tasks. This approach differs significantly from existing challenging benchmarks, which are either focused on complex tasks such as Big-Bench-Hard (Suzgun et al., 2022a) or specific to extremely few-shot settings like fewGLUE. In contrast to these previous studies (Hu et al., 2024), our proposed benchmark aims to generate difficult examples for any given task². In addition, real-world low-resource data samples often exhibit biases towards specific domains, such as blank backgrounds in image detection or short sentences in handwritten hate speech. Therefore, our evaluation also includes a bias assessment. Specifically, we consider two dimensions: misleading examples with smaller classification margins for performance evaluation, and biased examples for robust evaluation. We begin by conducting a comprehensive analysis of how these two dimensions impact low-resource learning. Based on the insights derived from our analysis, we present an empirical solution to construct a challenging low-resource benchmark. The final benchmark encompasses 3 computer vision datasets and 8 natural language processing datasets.

To prove the effectiveness of the constructed benchmark, we evaluate 13 models, including 8 pre-trained models, such as T5 (Raffel et al., 2020), Llama (Touvron et al., 2023a), etc. All these models struggle to handle our benchmarks, with a large performance gap compared with randomly-sampled low-resource benchmarks. On NLP tasks, we surprisingly find that despite better results on traditional low-resource benchmarks, pre-trained networks, do not show performance improvements on our benchmarks. The contribution of this paper is summarized as: **1)** We propose Achilles-Bench, a challenging benchmark designed to expose Achilles’ heel (weaknesses) of neural networks. This benchmark provides a reflective view of the current progress in the field of low-resource learning. **2)** We conduct a comprehensive analysis to identify the factors that particularly exacerbate the difficulty of low-resource learning. **3)**

²To ensure the exclusion of mislabelled examples, we have implemented a human-check process in our work.

Experimental results demonstrate that our proposed benchmark effectively challenges existing models, including robust pre-trained networks and large language models.

2 Related Work

Low-resource Evaluation Learning on low-resource datasets has recently come into the spotlight with the introduction of more powerful models (Radford et al., 2019; Brown et al., 2020). Recent low-resource benchmarks use a transfer learning setting (Dumoulin et al., 2021; Zheng et al., 2021) as well as in-context learning (Schick and Schütze, 2020; Bragg et al., 2021), and they have also added up on dataset difficulty (Wang et al., 2018). Among these, there are two major types of low-resource benchmark: natural low-resource datasets, and sampled low-resource datasets. The former requires additional dataset curation (Wang et al., 2018; Koh et al., 2021; Srivastava et al., 2022) and currently, most low-resource benchmarks are uniformly sampled from larger datasets (Kolesnikov et al., 2020; Schick and Schütze, 2020; Brown et al., 2020; Logan IV et al., 2021; Alayrac et al., 2022).

Challenging Benchmark Previous approaches in constructing challenging benchmark mainly curate from natural data (Schick and Schütze, 2020; Zheng et al., 2021; Xu et al., 2021; Koh et al., 2021). These methods require heavy annotation and faces misalignment between human-perceived difficulty and samples hard for models. Our methods, however, create an annotation-free framework for building challenging training sets, which has the potential to quickly apply to any available task. Other work involved benchmarking a more comprehensive and challenging list of tasks (Ye et al., 2021; Mukherjee et al., 2021; Hu et al., 2023b), which deviates from our focus in finding model weakness on common tasks.

Data Pruning Our approach is similar to data pruning literature in that we both hope to find a difficult subset in a large dataset. Previously, data pruning methods (Toneva et al., 2018; Hacoheh and Weinshall, 2019; Paul et al., 2021; Sorscher et al., 2022; Zhang et al., 2024) use data difficulty metrics including GradNorm and Loss Score to rank and prune datasets. However, we approach dataset sampling from a drastically different goal as we hope to challenge low-resource learning models.

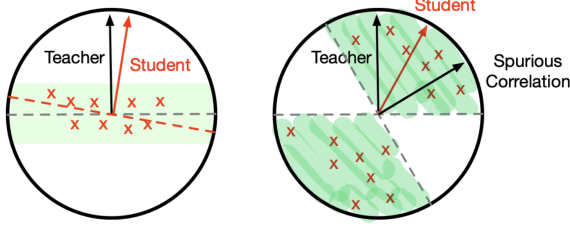


Figure 2: Plot of the perceptron model under hard low-resource learning (left) and biased low-resource learning setting (right). The green area shows the region where few-shot samples are sampled. (a) Under the hard low-resource learning setting, data samples are selected within a small margin to the decision boundary. (b) Under the biased low-resource learning setting, data samples are selected to satisfy the spurious classifier.

3 Understanding the Difficulty of Low-Resource Learning

To better understand the challenges of low resource learning, we first look at the teacher-student setting in learning perceptrons. Consider a large curated dataset of N examples $D = \{x_i, y_i\}_{i \in [N]}$ where $x_i \in \mathbb{R}^d$ are i.i.d. random Gaussian inputs $x_i \sim \mathcal{N}(0, I_d)$, with labels generated by a teacher perceptron $T \in \mathbb{R}^d$ as $y_i = \text{sign}(Tx_i)$. The number of samples $N \rightarrow \infty$ but sample per parameter $\alpha = \frac{N}{d} = O(1)$ to remain trainable. Now we consider the low resource scenario where the number of training samples available P is much less than N , where $\alpha_{\text{low}} = \frac{P}{d} \rightarrow 0$. For convenience, we sample the data for low resource learning from dataset D such that $D_{\text{low}} = \{x_\mu, y_\mu\}_{\mu \in [P]} \subset D$. Learning on D_{low} , we obtain a new student perceptron J that has generalization error ϵ_g .

Intuitively, three dimensions amount to the difficulty of learning perceptron J : (1) the number of training samples P (here we base the study of data scarcity on the sample per parameter variable α_{low}); (2) the classification difficulty of the data samples, denoted by the margin $m = \min_\mu J(x_\mu y_\mu)$; (3) the bias of the training dataset: here we look at a specific type of bias, spurious correlation, which draws correlation based on peripheral attributes of data items with a target variable, denoted as a student perceptron J_{bias} . We explore the difficulty of low-resource learning by altering our selection procedure for D_{low} and explore how ϵ_g changes. Specifically, we look at three settings and use simulation experiments for analysis. 1) Low-resource learning, where D_{low} is uniformly sampled from D . 2) Hard low-resource learning, where the margin of each sample is calculated $m_\mu = T(x_\mu y_\mu)$ and the samples with the smallest margins are se-

lected from D , as shown in Figure 2. 3) Biased low-resource learning, where a biased probe J_{bias} with θ angle to T is chosen as the spurious classifier. Then data that satisfies both $y_i = \text{sign}(J_{\text{bias}} x_i)$ and $y_i = \text{sign}(T x_i)$ is uniformly sampled from D , as shown in Figure 2.

We elaborate on simulation settings in the Appendix.

Difficult data especially challenges low resource learning. We first compare the setting that increases data difficulty to the random-sampled version of Low-resource Learning. We vary our dataset size from 1% to 500% trainable parameters. As shown in Figure 3, the dark blue line corresponds to the setting where data is uniformly selected, and lighter lines range in data difficulty from margin 0.1 to 1. The functions of ϵ_g to α yield a crossover between the function for random-sampled training data and the one for increased difficulty training data, showing that increased data difficulty affects low resource settings more than sufficient data settings. Also, the increase in generalization error is more distinct for slightly larger training sets. As when the low-resource training set only has a few samples, it requires model to have strong generalization ability to beat the rule of generalization $\epsilon \propto \alpha^{-1}$ and the task is challenging enough.

Low resource learning is more sensitive to spurious correlations. In the biased learning scenario as shown in Figure 4, we compare students trained on biased datasets (red lines) to students trained on random-sampled datasets (blue lines). When the bias probe is more distinct from the teacher (larger θ), the drop in performance is more distinct. This is in line with the phenomenon that when a model overfits on spurious features that contain information distant from semantics, the model tends to suffer on generalization. Also, for smaller bias, low resource learning sees a larger drop in generalization while models with abundant data barely suffer. This show that low-resource learning is sensitive to even small biases.

Theoretical perspective Here we use theoretical analysis in addition to simulations to study the scenario that results in failed generalization in low resource learning. Again, we focus on the scenario where we have a large dataset D that represents the natural task distribution P . We sample a low resource dataset D_{low} from D that form the distribution P_{low} . We theoretically show that the generalization error for the model trained on the low-

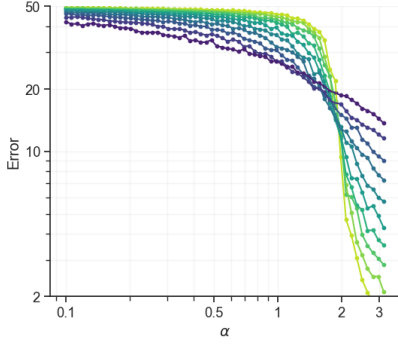


Figure 3: Plot of the generalization error with regard to data difficulty and the number of samples per parameter. Lighter lines represent more difficult data, and the dark blue line represents data uniformly selected.

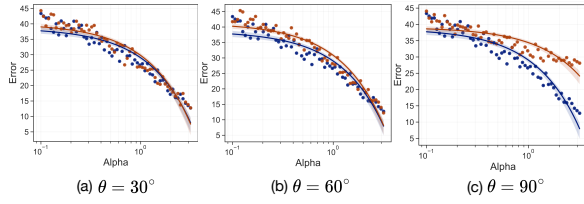


Figure 4: Plot of generalization error with regard to the number of samples per parameter. Red lines represent biased training set. Blue lines represent unbiased set.

resource dataset is bounded by a function of data difficulty and the distribution bias of low-resource dataset.

Theorem 3.1. (*Low-resource Generalization Measured by Distribution Shift and Data difficulty*) Let $\mathcal{H}$ be the hypothesis space $X \rightarrow \mathbb{R}^d$. f_{low} is the empirical risk $\epsilon_{P_{low}}(f)$ minimizer, and f is the hypothesis that minimizes expected risk $\epsilon_Q(f)$, m is the smallest margin of D to decision boundary of f . $MMD(P_{low}, P)$ describes the Maximum Mean Discrepancy (Gretton et al., 2012) between the sampled distribution and the original distribution. Then with probability over $1-\delta$,

$$\epsilon_Q(f_{low}) \leq \epsilon_Q(f) + c \sqrt{\frac{|\mathcal{H}| \ln m + \ln\left(\frac{2}{\delta}\right)}{m}} + MMD(P_{low}, P) + \epsilon_\alpha + \epsilon_{\mathcal{H}} \quad (1)$$

where ϵ_α , and $\epsilon_{\mathcal{H}}$ are small constants describing the error that occurred in training and the hypothesis space complexity, while c is the constant describing the scale of the effect of margin on generalization. Details are shown in Appendix.

The value of the Equation 1 right-hand side increases when m decreases and the term $MMD(P_{low}, P)$ increases, corresponding to the increase in data difficulty and the presence of data bias. This theorem applies not only to our simu-

lated scenario of perceptron learning but also to deeper models. In our biased learning setting, the distribution gap between low resource data distribution is larger for biased training set than random-sampled training set, i.e., $MMD(P_{low}^\theta, P) > MMD(P_{low}^{random}, P)$, since data samples forming P_{low}^{random} are sampled uniformly from P .

Based on our simulation experiments and theoretical results in the previous section, we find that low-resource learning is more likely to suffer from performance drop due to data difficulty and dataset bias. However, these scenarios are not covered in previous low-resource benchmarks. This motivates us to propose a challenging benchmark Achilles-Bench for better evaluation.

4 Achilles-Bench Challenge

We propose a new challenging benchmark that elevates low-resource learning difficulty on some well-known datasets. Unlike previous low-resource datasets that are randomly sampled from a training set, we curate the benchmark by selecting one of the most challenging low-resource training sets from GLUE, CIFAR10, CIFAR100, and ImageNet.

Following our theoretical analysis, we introduce the simple yet effective approach to build hard-Bench: First, we train a predictor for only one epoch on a large benchmark, obtaining a biased predictor; then, we score each sample on data difficulty for this stage of training. For each label, we pick the top k samples as our selected low-resource training set. We elaborate on the data difficulty metrics and the biased predictor respectively in section 4.1.

4.1 Metrics Measuring Data Difficulty

Previous literature in curriculum learning (Hacohen and Weinshall, 2019), data pruning (Paul et al., 2021), and continual learning (Toneva et al., 2018) propose metrics for data sample difficulty based on loss or gradient norms. Here we restate three metrics: *Loss score*, *GradNorm score* and explain how they can be applied in our problem scenario.

Loss Score Paul et al. (2021) and Sorscher et al. (2022) state this metric in the EL2N method, which intuitively measure data samples difficulty by looking at whether they can be learned correctly. Data samples with a higher loss score after training are more likely to be near the decision boundary. Therefore, we can select the hardest samples by ranking the loss score on the dataset. We call

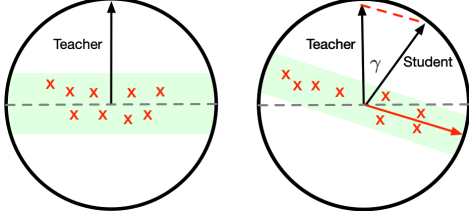


Figure 5: Plot of the perceptron model under both hard and biased low-resource learning setting. Compared to the no-bias setting on the left, the resulting bias is $\frac{\gamma}{2}$ when the gap between J_{bias} and teacher is γ .

datasets constructed via loss scores as **Achilles-Bench (Loss)**. Examples with higher losses are selected as hard examples.

Gradient Norm Score Paul et al. (2021) discussed using gradient norm as an indicator of data importance. Samples with larger gradient norms shape the training geometry. However, there is little discussion on the connection between gradient norm and data difficulty. Here we give a brief and casual explanation. Based on previous analysis, we can find hard samples by checking their margin to the decision boundary of our model f , $f(x_0) = 0$. Therefore, we can define the L_p norm margin as,

$$m(x) = \min_{x_0} \|x - x_0\|_p, s.t. f(x_0) = 0 \quad (2)$$

We use Taylor’s approximation for an approximate solution, following Elsayed et al. (2018).

$$m(x) \approx \frac{|f(x)|}{\|\nabla_x f(x)\|_q}, \quad (3)$$

When the numerator is constrained (For a classification problem, we can constraint logits $f(x)$ within 1 using sigmoid function), we can maximize the gradient norm to minimize margin. We call datasets constructed via gradient norm scores as **Achilles-Bench (GradNorm)**. Examples with higher gradient norm scores are selected as hard examples.

4.2 Introducing Bias with Early Stopping

As shown in the above sections, we need to train a student predictor to estimate the decision boundary and thereby calculate the data difficulty score. However, we find that we can easily introduce bias into our selected benchmark dataset if we early stop training on the student predictor. We will give an explanation based on the Loss Score.

The Loss Score effectively estimates the difficulty of data examples to be classified correctly when the student predictor is exactly the same as the teacher model, i.e. $\theta = 0$. However,

when the student model is undertrained, there would exist a gap γ between student $g(x) = \text{sgn}(Jx)$ and teacher $f(x) = \text{sgn}(Tx)$. For any x , the loss function would be $L(x) = g(x) - f(x) = (J - T)x$. Therefore, the resulting selected dataset $D_{low} = \{(x_i, y_i) | x_i = \max_{i=1,2,\dots,P} (J - T)x, y_i = \text{sgn}(Tx_i)\}$ is isotropic in the nullspace of $J - T$, inducing a bias of $\frac{\gamma}{2}$.

This intuitively explains that we can use an early stopped predictor as well as data difficulty metrics to select a biased and difficult low-resource dataset that mimics the real-world setting. In the following sections, we use this approach to curate our Achilles-Bench.

5 Experiments

5.1 Benchmark Metric

Traditional low-resource benchmarks usually randomly choose a subset from the full-size training data as the training set. In this paper, we also follow this setting and extract hard examples from the full-size data as the training data in our benchmark. To be specific, we implement three benchmarks in this work, which are described as follows. **Random-Bench**. For each label, we randomly select k examples as the training set. We randomly select 3 subsets and report the average results. **Achilles-Bench (Loss)**. For each label, we choose top- k hard examples based on losses scores. **Achilles-Bench (GradNorm)**. For each label, we choose top- k hard examples based on gradient norm scores.

5.2 Benchmark Settings

Our framework is not limited to specific tasks, allowing for flexibility across various tasks. We benchmark on from-scratch models, pre-trained models, as well as large language models. In our implementation, we have chosen 11 tasks to generate a comprehensive and challenging benchmark.

NLP Tasks We choose 8 datasets from GLUE (Wang et al., 2018), a collection of understanding datasets. We select a subset of the full-size training set as a training set. Following previous studies, we use the validation set as the test set considering the hidden test set. For the convenience of the demonstration, we show all the results with accuracy scores. For all NLP datasets, we implement BERT trained with one epoch as a biased predictor to select hard examples. For all NLP datasets, we extract 500 examples for each label (except for WNLI with 100 examples) as

Models	SST-2	COLA	MNLI	QNLI	MRPC	QQP	RTE	WNLI	Average
<i>Random-Bench</i>									
Transformer (Vaswani et al., 2017)	68.16±1.46	69.15±0.04	36.42±0.58	55.45±0.94	68.58±0.39	67.18±0.67	53.65±1.04	56.34±0.00	59.37
BERT (Devlin et al., 2018)	88.68±0.73	79.00±0.59	57.60±1.30	76.02±1.26	77.65±1.38	75.53±0.48	60.58±2.01	48.45 ±3.63	70.44
GPT-2 (Radford et al., 2019)	88.08±0.72	70.35±1.76	58.35±1.65	74.11±2.56	75.93±0.47	76.22±0.86	65.49±2.62	56.90 ±3.40	70.68
RoBERTa (Liu et al., 2019)	91.54 ±0.61	80.98 ±0.56	75.40 ±0.52	84.47 ±0.53	88.24 ±0.27	80.93 ±0.56	73.00 ±1.98	54.93±2.82	78.69
T5 (Raffel et al., 2020)	88.73±0.97	78.62±0.58	64.53±2.48	82.56±0.83	74.56±1.71	80.13±0.44	56.46±1.95	52.39±7.74	72.25
<i>Achilles-Bench (GradNorm)</i>									
Transformer (Vaswani et al., 2017)	51.88±0.46	69.15±0.04	35.11±0.67	50.59±0.04	68.38±0.00	62.41±1.06	54.01±0.96	56.34±0.00	55.98
BERT (Devlin et al., 2018)	47.94±2.11	45.77±8.19	33.96±0.47	46.24±2.35	56.08±1.43	52.60±3.01	51.12±0.96	49.30±1.99	47.88
GPT-2 (Radford et al., 2019)	51.44±0.77	51.93±7.92	35.98±1.95	48.62±5.12	65.98±2.33	55.40±4.05	57.76±4.60	56.06±2.25	52.90
RoBERTa (Liu et al., 2019)	51.01±0.65	66.10±6.01	38.42±1.51	48.61±1.50	82.55±1.04	56.69±3.93	60.36±2.88	54.93±2.18	
T5 (Raffel et al., 2020)	52.34±2.35	55.09±6.16	34.27±0.39	48.99±1.51	55.88±3.80	55.72±1.62	48.88±1.54	54.37±4.14	50.69
<i>Achilles-Bench (Loss)</i>									
Transformer (Vaswani et al., 2017)	51.38±0.40	69.11±0.04	34.98±0.69	50.57±0.04	65.64±5.49	48.17±7.69	53.43±0.40	56.34±0.00	53.70
BERT (Devlin et al., 2018)	45.64 ±5.32	40.92 ±4.29	30.55 ±0.88	40.11 ±3.69	38.24±2.52	35.55±2.57	47.44 ±1.22	53.52±3.67	41.50
GPT-2 (Radford et al., 2019)	49.79±2.06	56.18±9.92	31.41±1.19	51.01±3.89	50.54±8.02	40.33±5.57	54.73±3.67	55.49±1.44	48.69
RoBERTa (Liu et al., 2019)	50.55±0.62	48.32±11.78	31.66±2.49	41.79±5.62	38.14 ±2.54	31.74 ±2.44	55.09±1.97	55.77±1.91	44.13
T5 (Raffel et al., 2020)	49.86±2.85	55.32±6.06	32.76±0.23	47.15±1.76	53.19±5.12	48.84±5.38	48.45±1.20	53.52±4.45	48.64

Table 1: Results on NLP datasets. Achilles-Bench (Loss) brings higher performance drops than Achilles-Bench (GradNorm). Surprisingly, pre-trained networks does not show better generalization results than randomly-initialized models on our benchmark.

the training set for our main results. Regarding large language models, we adopted the in-context learning paradigm, details can be find in Appendix E. We also build more variants with less training data. More results can be found at Appendix F.

CV Tasks We also explore 3 widely-used image classification datasets, CIFAR-10 (Krizhevsky et al., 2009), CIFAR-100 (Krizhevsky et al., 2009), and ILSVRC-2012 ImageNet (Deng et al., 2009) to demonstrate the generality of our approach. For each dataset, we select a subset as the training set in our benchmark, with 500 examples in CIFAR-10, 50 examples in CIFAR-100, 100 examples in ImageNet-1K. results can be found at Appendix F.

5.3 Results

Achilles-Bench challenges neural networks As Table 1, Table 2 and Table 3 illustrate, Achilles-Bench can mislead neural networks with worse generalization errors. We re-implement strong understanding models, which have shown promising results in various low-resource tasks. For example, in Random-Bench, RoBERTa shows the near-human performance on SST-2 with 91% accuracy, which drops sharply on Achilles-Bench with only 51.01% accuracy on Achilles-Bench (GradNorm) and 50.55% accuracy on Achilles-Bench (Loss), nearly random-guessing results. Similar results are observed on CV datasets. For example, DenseNet-121 trained on a random sampling set achieves high test results with 71.33% accuracy on CIFAR-10. The accuracy drops to 59.87% on Achilles-Bench (GradNorm) and to 44.81% on Achilles-Bench (Loss). For LLMs, LLaMA-7B and LLaMA2-7B consistently demonstrate the low-

est performance on Achilles-Bench. Regarding BLOOM-1.1B’s performance on QQP, it is noteworthy that the model’s results are subpar compared to the label distribution, where "not duplicate" constitutes 63.2% of the dataset. The large performance drop also indicates that there is still a large gap between existing models and human-level models. All these drops demonstrate that our benchmark poses a great challenge.

Pre-trained networks show strong generation results on CV benchmarks, but still suffer from handling NLP tasks Compared with randomly-initialized models, pre-trained networks show better generalization results in CV datasets, as shown in Table 3. For example, ViT-B/16 does not yield obvious performance drops on Achilles-Bench. As a comparison, pre-trained networks have much worse results on NLP tasks. On Random-Bench, pre-trained networks bring large performance improvements over random-initialized baseline (Transformer). However, on our benchmark, all pre-trained networks yield surprising performance drops. These results demonstrate that the results of pre-trained models on NLP tasks are more easily over-estimated.

Achilles-Bench (Loss) is more challenging than Achilles-Bench (GradNorm) We implement two metrics to select hard examples, including loss and gradient norm. Despite similar motivation, Achilles-Bench (loss) is more challenging than Achilles-Bench (GradNorm) according to our experimental results. On NLP tasks, Achilles-Bench (loss) also witnesses the worst results. Loss is the

Models	SST-2	COLA	MNLI	QNLI	MRPC	QQP	RTE	WNLI	Average
<i>Random-Bench</i>									
BLOOM-1.1B (Scao et al., 2022)	50.5	60.4	35.4	50.5	66.2	51.8	52.7	42.3	51.2
Llama-7B (Touvron et al., 2023a)	60.2	63.1	33.1	48.3	67.4	47.9	51.0	47.9	52.4
Llama2-7B (Touvron et al., 2023b)	95.4	68.9	53.7	58.0	68.1	73.7	79.4	63.4	68.5
Llama2-13B (Touvron et al., 2023b)	85.1	80.5	49.5	54.9	70.5	78.1	75.3	68.5	70.3
Llama2-70B (Touvron et al., 2023b)	90.3	78.8	61.7	49.8	68.4	42.4	79.2	85.5	69.5
<i>Achilles-Bench (Loss)</i>									
BLOOM-1.1B (Scao et al., 2022)	50.1	46.4	35.42	50.0	65.9	60.8	47.3	43.7	50.0
Llama-7B (Touvron et al., 2023a)	40.7	61.4	30.6	46.3	68.1	40.4	49.1	42.3	47.4
Llama2-7B (Touvron et al., 2023b)	64.6	53.2	46.4	59.7	68.1	79.5	76.5	64.8	63.0
Llama2-13B (Touvron et al., 2023b)	48.4	78.6	43.0	47.4	69.6	76.8	74.7	66.2	63.1
Llama2-70B (Touvron et al., 2023b)	48.4	71.0	43.8	47.0	68.4	37.2	76.1	90.1	60.3

Table 2: The in-context learning results of LLMs on NLP datasets. Achilles-Bench (Loss) consistently preserve its challenges for LLMs.

most direct signal to see how neural networks understand an example. These difficult examples confuse neural networks, which barely learn core features. This learning weakness is not covered by existing low-resource benchmarks. Achilles-Bench provides a new perspective for understanding the learning abilities of different models.

Data augmentation slightly improves results

Table 4 shows the results on CIFAR-10 with data augmentation techniques, cutmix (Yun et al., 2019). We can see that data augmentation brings slight performance improvements, but also faces the challenges of generalization on our benchmarks.

Models	CIFAR10	CIFAR100	ImageNet
<i>Random-Bench</i>			
FFN	48.91±0.87	14.95±0.29	5.12±0.30
VGG-16	62.15±0.71	26.55±0.20	16.02±0.27
ResNet-18	65.47±0.84	25.49±0.60	29.34±0.31
DenseNet-121	71.33±0.56	33.66±1.48	35.20 ±0.41
ViT-B/16	97.20±0.22	83.93 ±0.43	-
EfficientNetV2-S	91.41±0.60	70.41±0.74	-
<i>Achilles-Bench (GradNorm)</i>			
FFN	29.64±0.88	8.75±0.28	3.13±0.18
VGG-16	55.11±0.89	17.22±0.44	9.51±0.20
ResNet-18	46.87±2.41	15.50±0.85	23.81±0.76
DenseNet-121	59.87±0.66	20.96±0.94	28.96±0.67
ViT-B/16	97.39 ±0.10	82.36±0.94	-
EfficientNetV2-S	92.51±0.24	69.56±0.49	-
<i>Achilles-Bench (Loss)</i>			
FFN	17.26 ±0.82	3.18 ±0.21	2.66 ±0.02
VGG-16	27.58±0.62	7.14±0.24	7.27±0.24
ResNet-18	33.20±1.00	6.96±0.32	13.34±0.19
DenseNet-121	44.81±2.30	11.59±0.98	22.00±0.46
ViT-B/16	96.85±0.11	80.87±0.58	-
EfficientNetV2-S	89.88±0.63	60.42±1.85	-

Table 3: Results on CV datasets. ViT and efficientNetV2-S are pre-trained on ImageNet. So we do not report their results on ImageNet to avoid data leak issues.

Models	Random-Bench	Achilles-Bench (GradNorm)	Achilles-Bench (Loss)
FFN	53.99±0.39	30.36±1.26	19.29±0.36
VGG-16	66.76±0.59	47.85±0.97	33.64±0.25
ResNet-18	68.94±0.66	52.73±1.54	37.96±1.12
DenseNet-121	75.44±0.34	63.23±0.42	47.70±1.38
ViT-B/16	97.71±0.17	97.79±0.08	97.09±0.12
EfficientNetV2-S	93.25±0.65	92.83±0.63	91.41±0.69

Table 4: Results with cutmix. Models with data augmentation still face the challenges of generalization on our benchmarks.

Models	Random-Bench	Achilles-Bench (Loss)	FewGLUE
RoBERTa	57.8± 3.62	52.0	62.8
GPT-2	58.8± 2.65	47.3	47.7

Table 5: Results compared with FewGLUE on 32-shot RTE.

Models	Achilles-Bench (Loss)		Forget Statistic	
	Accuracy	Gap	Accuracy	Gap
FFN	16.17	30.33	33.11	13.39
VGG-16	26.78	33.03	43.00	16.81
ResNet-18	32.10	30.64	45.77	16.97
DenseNet-121	41.45	28.80	59.63	10.62
ViT	96.70	0.25	97.48	-0.53
EfficientNet-V2	89.17	0.54	91.10	-1.39

Table 6: The comparison between Achilles-Bench (Loss) and Forget Statistic on CIFAR-10. “Gap” represents the test accuracy gap with Bench-Random.

Achilles-Bench (Loss) demonstrate greater challenges compared to FewGLUE (Schick and Schütze, 2020) Table 5 presents a performance comparison between RoBERTa and GPT-2 on the 32-shot RTE task. The performance of GPT-2 under both the Achilles-Bench (Loss) and FewGLUE approaches tends to resemble random selection. Regarding RoBERTa, FewGLUE does not seem significantly more challenging than Random-Bench, whereas Achilles-Bench (Loss) demonstrates a higher level of difficulty.

Results on different metrics Table 6 presents the outcomes obtained on the 500-shot datasets from CIFAR-10 using the forget statistic technique (Toneva et al., 2018). Achilles-Bench (Loss) surpasses the forget statistic approach in all models,

including pre-trained models. The forget statistic technique does not appear to be more challenging than Random-Bench for pre-trained models.

5.4 Ablation Studies

Massive sampling fails to find a challenging benchmark In Random-Bench, we report the average results over 3 random samplings. In this part, we conduct 100 samplings and report the worst result in Figure 6 to figure out whether our methods can be replaced with massive sampling. As we can see, there is still a large gap between the worst results on Random-Bench and Achilles-Bench, indicating that the proposed method is an effective method to build challenging benchmarks.

Results on the selected set as the test set Figure 7 shows results on the selected set as the test set. As we can see, these “hard examples” capture the weakness of neural networks. If neural networks has not seen these examples, they fail on them.

Ablation studies on different models as predictors In our framework, we introduce a weak classifier as a biased predictor. For simplification, we choose FFN for CV datasets and BERT for NLP datasets. We conduct experiments on more networks to see whether the choice of predictors affects our conclusions. Table 7 and Table 8 show the attack results on SST-2 and CIFAR-10. For SST-2, we test two more models: randomly-initialized Transformer and GPT2, as predictors. For CV models, we test two more models: ResNet-18 and ViT-B/16, as predictors. All models show consistent

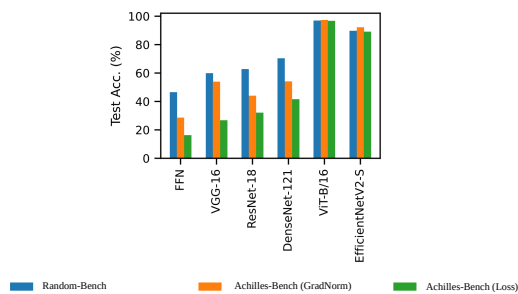


Figure 6: The worst performances among all the performances on CIFAR-10.

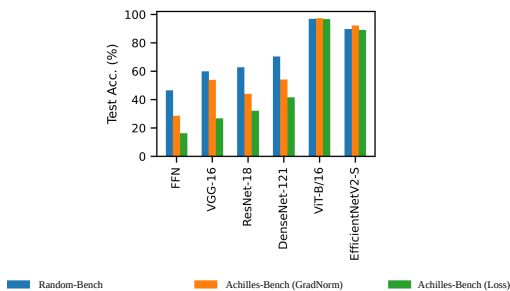


Figure 7: Results on the selected set as the test set.

performance drops, indicating that our method is a universal model to generate challenging datasets to attack various models.

Models	Transformer Predictor		GPT-2 Predictor	
	Accuracy	Gap	Accuracy	Gap
Transformer	51.17± 0.17	16.99	50.55± 0.73	17.61
BERT	51.06± 2.51	37.62	48.30± 2.00	40.38
GPT-2	50.46± 3.20	37.62	48.88± 4.00	39.20
RoBERTa	54.72± 3.04	36.82	48.33± 2.19	43.21
T5	60.48± 3.88	28.25	56.03± 2.62	32.70

Table 7: Results of Achilles-Bench (Loss) on SST-2 based on a random initialized Transformer and GPT-2. “Gap” represents the test accuracy gap with Bench-Random.

Models	ResNet Predictor		ViT Predictor	
	Accuracy	Gap	Accuracy	Gap
FFN	40.69± 0.69	8.32	39.82± 0.61	9.19
VGG-16	51.83± 0.39	16.80	48.19± 0.64	20.44
ResNet-18	53.93± 0.72	11.58	50.59± 0.98	14.92
DenseNet-121	61.70± 0.23	9.72	58.05± 0.80	13.37
ViT-B/16	97.07± 0.19	0.00	96.92± 0.32	0.15
EfficientNet-V	89.70± 0.32	2.12	87.26± 1.01	4.56

Table 8: Results of Achilles-Bench (Loss) on CIFAR10 based on ResNet-18 and ViT-B/16. “Gap” represents the test accuracy gap with Bench-Random.

5.5 Explaining the Effectiveness of Achilles-Bench with Visualization

In this section, we compare samples selected by our Achilles-Bench with samples from Random-Bench to demonstrate our approach reaches the goal of building difficult low-resource training set with shifted distributions. To make our observation more straightforward, we show visualizations in the Appendix G. We make the following observations based on these visualization results:

Achilles-Bench induces bias in the low-resource training set

From visualizations, we can see that both GradNorm and Loss variations of Achilles-Bench construct training sets that are drastically different from the data distribution. For SST2 task, specifically, Random-Bench exhibits ordinary statements containing words with clear emotional expressions. In contrast, both GradNorm and the Loss variations of Achilles-Bench opt for shorter sentences, incorporating statements with implicit emotional nuances. Similar biases are evident in other classes, showing that our approach successfully induces bias in the low-resource training set.

Achilles-Bench find challenging samples

The method selects tough examples from datasets using difficulty metrics, notably in GradNorm and Loss. In the SST2 task, it favors terse, uninformative samples or input sentences that use sophisticated

vocabularies.

6 Conclusion

This paper proposes a challenging benchmark for low-resource learning. We first analyze which factors affect the difficulty of low-resource learning. We prove that low-resource generalization results in worse performance with more difficult and biased datasets. Hence we choose two metrics for measuring data difficulty, which result in two variants, Achilles-Bench (Loss) and Achilles-Bench (GradNorm). Experiments show that both can better tell the learning gap between existing models than randomly-sampled low-resource datasets.

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A Limitation and Future Work

The method proposed in this paper can be extended to a wider range of tasks and datasets. In future studies, we aim to expand the tasks to more challenging datasets, such as superGLUE.

Furthermore, we did not test the results on the latest models, such as GPT-4. We intend to extend the tasks to include the latest large-scale models in the future.

B Perceptron Model of Low Resource Learning

In this section, notations are defined as follows. We look at the teacher-student setting in learning perceptrons. Consider a large curated dataset of N examples $D = \{x_i, y_i\}_{i \in [N]}$ where $x_i \in \mathbb{R}^d$ are i.i.d. random Gaussian inputs $x_i \sim \mathcal{N}(0, I_d)$, with labels generated by a teacher perceptron $T \in \mathbb{R}^d$ as $y_i = \text{sign}(Tx_i)$. The number of samples $N \rightarrow \infty$ but sample per parameter $\alpha = \frac{N}{d} = O(1)$ to remain trainable. Now we consider the low resource scenario where the number of training samples available P is much less than N , where $\alpha_{\text{low}} = \frac{P}{d} \rightarrow 0$. For convenience, we sample the data for low resource learning from dataset D such that $D_{\text{low}} = \{x_\mu, y_\mu\}_{\mu \in [P]} \subset D$. Learning on D_{low} , we obtain a new student perceptron J that has generalization error ϵ_g .

In the basic low-resource learning scenario, we use a uniform sampling strategy to obtain D_{low} from D . We model ϵ_g as a function of α_{low} . The results are as follows.

Lemma B.1. (*Low-resource Learning, Seung et al. (1992)*) For student perceptron J learned on high dimension dataset D_{low} , the generalization error satisfies,

$$\epsilon_g \propto \alpha_{\text{low}}^{-1} \quad (4)$$

For other settings, the generalization error is only related to the angle between teacher model T and learned student model J . $\epsilon_g = \arccos R / \pi$, $R = \frac{JT}{\|J\|\|T\|}$. Based on different low-resource dataset sampling strategies, we calculate the teacher-student overlap R with the geometry of each dataset distribution. Sorscher et al. (2022) has proved similar results in data pruning with our hard and biased learning setting. Here we only cite their results and don't elaborate on proofs. Note that despite the proofs being similar, we use a different setting in perceptron learning. Their main objective is to understand how data-pruning can improve data efficiency, while we take an inverse stand, trying to understand challenging settings of low-resource learning.

Lemma B.2. (*Hard Low-resource Learning, Sorscher et al. (2022)*) D_{low} is sampled from D such that $\forall x_\mu \in D_{\text{low}}, \forall x_\gamma \in D/D_{\text{low}}$, their margins satisfy $|Tx_\mu| \geq |Tx_\gamma|$. Let J be the student perceptron learned on high dimension dataset D_{low} , and κ be the minimum margin $\min_\mu J(x_\mu y_\mu)$. If the perceptron is trained to maximum margin, the generalization error of J satisfies,

$$\epsilon_g = \arccos R / \pi \quad (5)$$

where R satisfies the saddle point equation,

$$R = \frac{2\alpha}{f\sqrt{2\pi}\sqrt{1-R^2}} \int_{-\infty}^{\kappa} Dt \exp\left(-\frac{R^2 t^2}{2(1-R^2)}\right) \cdot \left[1 - \exp\left(-\frac{\gamma(\gamma - 2Rt)}{2(1-R^2)}\right)\right] (\kappa - t) \quad (6)$$

in which $\gamma = H^{-1}(\frac{N-P}{2N})$, $p(z) = \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi P}} \Theta(\gamma - |z|)$

The proof for this lemma can be found in Sorscher et al. (2022) A.5.1 and is omitted here for brevity.

C Low-Resource Generalization

C.1 Proof for Theorem 3.1

Lemma C.1. Define $\epsilon_S(h, f) := E_{x \sim S} |\delta(h(x)) - \delta(f(x))|$. For any hypothesis $h, h' \in \mathcal{H}$, there exists $\epsilon_H > 0$ which satisfies,

$$|\epsilon_{P_{low}}(h, h') - \epsilon_P(h, h')| \leq \text{MMD}(\mathcal{H}, P_{low}, P) + \frac{\epsilon_{\mathcal{H}}}{2} \quad (7)$$

ϵ_H is a constant for the complexity of hypothesis space.

Lemma C.2. Let f_{low} be the trained classifier on the low resource distribution P_{low} , and f be the trained classifier on distribution P . Since P_{low} is formed by a subset of the training examples, when training error $\epsilon_P(f) \rightarrow 0$ and $\epsilon_{P_{low}}(f_{low}) \rightarrow 0$, $\epsilon_{P_{low}}(f_{low}, f) \leq \epsilon_{\alpha}$, where ϵ_{α} is a constant approaching zero.

Proof.

$$\begin{aligned} |\epsilon_{P_{I_k}}(h, h') - \epsilon_P(h, h')| &\leq \sup_{h, h' \in \mathcal{H}} |\epsilon_{P_{I_k}}(h, h') - \epsilon_P(h, h')| \\ &= \sup_{h, h' \in \mathcal{H}} |\mathbf{P}_{\mathbf{x} \sim P_{I_k}}[\delta(h(\mathbf{x})) \neq \delta(h'(\mathbf{x}))] - \mathbf{P}_{\mathbf{x} \sim P}[\delta(h(\mathbf{x})) \neq \delta(h'(\mathbf{x}))]| \\ &= \sup_{h, h' \in \mathcal{H}} |\mathbf{P}_{\mathbf{x} \sim P_{I_k}}[h(\mathbf{x}) \neq h'(\mathbf{x})] - \mathbf{P}_{\mathbf{x} \sim P}[h(\mathbf{x}) \neq h'(\mathbf{x})]| \\ &= \sup_{h, h' \in \mathcal{H}} \left| \int_{\mathcal{X}} \mathbf{1}_{h(\mathbf{x}) \neq h'(\mathbf{x})} d\mu_{P_{I_k}} - \int_{\mathcal{X}} \mathbf{1}_{h(\mathbf{x}) \neq h'(\mathbf{x})} d\mu_P \right| \end{aligned} \quad (8)$$

Lemma C.3. Let f be the trained classifier on dataset D that is drawn i.i.d. from distribution P . f is tested on a test dataset S that is also drawn i.i.d. from the distribution P . Let m be the maximum margin of classifier f . Then with probability at least $1 - \delta$,

$$\epsilon_S(f) \leq \epsilon_D(f) + c \sqrt{\frac{|\mathcal{H}| \ln m + \ln(\frac{1}{\delta})}{m}} \quad (9)$$

where $\epsilon_D(f)$ is the error on training set, and $\epsilon_S(f)$ be the error on test set.

Following Ben-David et al. (2010), we use Lemma C.1 and C.2 to prove Theorem 3.1.

Proof

$$\begin{aligned} \epsilon_Q(f_{low}) &\leq \epsilon_Q(f) + \epsilon_Q(f_{low}, f) \\ &= \epsilon_Q(f) + \epsilon_{P_{low}}(f_{low}, f) + (\epsilon_Q(f_{low}, f) - \epsilon_P(f_{low}, f)) + (\epsilon_P(f_{low}, f) - \epsilon_{P_{low}}(f_{low}, f)) \\ &\leq \epsilon_Q(f) + \epsilon_{P_{low}}(f_{low}, f) + |\epsilon_P(f_{low}, f) - \epsilon_Q(f_{low}, f)| + |\epsilon_{P_{low}}(f_{low}, f) - \epsilon_P(f_{low}, f)| \\ &\leq \epsilon_Q(f) + \epsilon_{\alpha} + |\epsilon_P(f_{low}, f) - \epsilon_Q(f_{low}, f)| + \text{MMD}(P_{low}, P) + \epsilon_{\mathcal{H}} \end{aligned} \quad (10)$$

In which,

$$\begin{aligned} |\epsilon_P(f_{low}, f) - \epsilon_Q(f_{low}, f)| &= \left| \int_{\mathcal{X}} \mathbf{1}_{f_{low}(\mathbf{x}) \neq f(\mathbf{x})} d\mu_P - \int_{\mathcal{X}} \mathbf{1}_{f_{low}(\mathbf{x}) \neq f(\mathbf{x})} d\mu_Q \right| \\ &= \left| \sum_{i=1}^n \mathbf{1}_{f_{low}(\mathbf{x}_i) \neq f(\mathbf{x}_i)} - E_Q \mathbf{1}_{f_{low}(\mathbf{x}) \neq f(\mathbf{x})} \right| \end{aligned} \quad (11)$$

Here suppose test set Q matches the distribution of data for this classification task, and P is constructed by sampling n i.i.d. samples from the distribution Q . Using Lemma C.3 we have,

$$P(|\epsilon_P(f_{low}, f) - \epsilon_Q(f_{low}, f)| > c \sqrt{\frac{|\mathcal{H}| \ln m + \ln(\frac{2}{\delta})}{m}}) \leq \delta \quad (12)$$

Therefore, with a probability over $1 - \delta$, we have

$$\epsilon_Q(f_{low}) \leq \epsilon_Q(f) + \text{MMD}(P_{low}, P) + \epsilon_{\alpha} + \epsilon_{\mathcal{H}} + c \sqrt{\frac{|\mathcal{H}| \ln m + \ln(\frac{2}{\delta})}{m}} \quad (13)$$

□

Table 9: Hyper-parameter settings. The Linear refers LinearLR scheduler in Pytorch. OneCycle refers 1-cycle learning rate policy (Smith and Topin, 2019).

Models	Datasets	Batch Size	Epochs	Optimizer	Learning Rate
FFN	CIFAR-10	128	50	Adam	[1e-3, 5e-4, 2.5e-4]
	CIFAR-100	128	50	Adam	[1e-3, 5e-4, 2.5e-4]
	ImageNet-1K	32	30	SGD	[0.01, 0.001, 0.0001]
VGG	CIFAR-10	128	50	Adam	[1e-4, 5e-5, 2.5e-5]
	CIFAR-100	128	50	Adam	[1e-4, 5e-5, 2.5e-5]
	ImageNet-1K	32	30	SGD	[0.01, 0.001, 0.0001]
ResNet	CIFAR-10	128	50	Adam	[1e-3, 5e-4, 2.5e-4]
	CIFAR-100	128	50	Adam	[1e-3, 5e-4, 2.5e-4]
	ImageNet-1K	32	30	SGD	[0.1, 0.01, 0.001]
DenseNet	CIFAR-10	128	50	Adam	[1e-3, 5e-4, 2.5e-4]
	CIFAR-100	128	50	Adam	[1e-3, 5e-4, 2.5e-4]
	ImageNet-1K	32	30	SGD	[0.1, 0.01, 0.001]
ViT-B/16	CIFAR-10	32	10	Adam	5e-5 (Linear)
	CIFAR-100	32	10	Adam	5e-5 (Linear)
EfficientNetV2-S	CIFAR-10	32	10	AdamW	1e-3 (OneCycle)
	CIFAR-100	32	10	AdamW	1e-3 (OneCycle)

D Models and Hyperparameters

We implement the following models for experiments in this paper.

1) **FFN**, a feed-forward neural network with two convolution and pooling layers and three feed-forward layers. 2) **VGG** (Simonyan and Zisserman, 2014), a classical convolutional neural network. We use the VGG-16 with 13 convolution layers and three fully connected layers as implementation. 3) **ResNet** (He et al., 2016), a residual neural network. We use the ResNet-18 with 16 residual blocks, one convolution layer, and one fully connected layer as implementation. 4) **DenseNet** (Huang et al., 2017). We use DenseNet-121 with 121 layers, one convolution layer, and one fully connected layer as re-implementation.³ Besides, to verify the attack ability Gradon the pre-trained models, we also re-implement two pre-trained models: 1) Transformer-based **ViT** (Dosovitskiy et al., 2021)⁴ and 2) Convolutional-based **EfficientNetV2** (Tan and Le, 2021)⁵. For FFN, VGG, ResNet, ResNeXt, and DenseNet on ImageNet, we resize all the images into 256×256 and then center-crop them into 224×224 . For ViT on CIFAR, we resize all the images into 224×224 , while 384×384 for EfficientNetV2.

We list hyper-paramters in Table 9. All the SGD optimizers are with a momentum of 0.9. For Adam/AdamW, we set $\beta = (0.9, 0.999)$. We employed the torch.optim.lr_scheduler.MultiStepLR module to dynamically adjust the learning rate during training. Specifically, we set the milestones at epochs 20 and 40 (for ImageNet-1K, epochs 10 and 20 respectively) to adaptively update the learning rate based on the progress of training. The corresponding learning rate values used during three periods in our experiments are provided in Table 9. We conduct all the experiments on a single A100 GPU. We use Adam as the optimizer for all the NLP tasks with a learning rate of $2e - 5$ and a linear scheduler.

³For VGG, ResNet, ResNeXt, and DenseNet on CIFAR and MNIST, we use the implementation from <https://github.com/kuangliu/pytorch-cifar>. As for ImageNet, we use the implementation from torch.models.

⁴We use the implementation from <https://huggingface.co/google/vit-base-patch16-224>

⁵We use the implementation from torch.models.

E Details of ICL and LLMs

Considering the sentence length of different tasks and limitations of the GPU, we tested SST2 and COLA with 16-shots, MNLI, QNLI, MRPC, RTE, WNLI with 8-shots each, and QQP with 4-shots.

We generate the prompt refer to lm-eval, the results of the prompt with zero-shots are shown below.

Models	SST-2	COLA	MNLI	QNLI	MRPC	QQP	RTE	WNLI	Average
Llama2-7B	86.70	38.70	50.10	62.00	58.09	63.30	72.56	61.97	61.68

Table 10: The zero-shot results of LLaMA2 on NLP datasets.

F Results with Different Shots

To better explore the effectiveness of Achilles-Bench (GradNorm) and Achilles-Bench (Loss) , we demonstrate the results with different shots. The results are shown in the following tables.

Based on the results presented in Table 11 and Table 12, as compared to the findings discussed in Section 5.4, it is evident that all the model demonstrates a notable decline in performance when trained on a more limited dataset (20, 50-shot) in Achilles-Bench (Loss) and Achilles-Bench (GradNorm) , as compared to Random-Bench . This observation suggests that Achilles-Bench (Loss) and Achilles-Bench (GradNorm) pose greater challenges with fewer shots. While the pretrained model exhibits some level of robustness in Section 5.4, its performance still suffers when faced with more limited data. This highlights the significance of employing challenging benchmarks that incorporate scenarios with limited training data.

The results in NLP, as shown in Table 14, Table 15, Table 16, are generally consistent with the previous findings presented in Section 5.4. Nonetheless, a few specific models, characterized by inadequate few-shot learning capabilities, exhibited poor performance across all three benchmarks.

Table 11: Results on CIFAR-10.

shots	Models	Random-Bench Accuracy	Achilles-Bench (GradNorm) Accuracy	Gap	Achilles-Bench (Loss) Accuracy	Gap
20-shot	FFN	27.28± 1.51	13.68± 0.57	13.60	10.74± 1.38	16.54
	VGG-16	31.73± 1.26	14.76± 0.86	16.97	10.27± 0.32	21.46
	ResNet-18	30.54± 1.82	14.80± 0.56	15.74	10.79± 0.35	19.75
	DenseNet-121	34.69± 1.51	15.25± 0.29	19.44	10.15± 0.61	24.54
	ViT-B/16	79.84± 1.70	62.92± 1.97	16.92	57.62± 2.46	22.22
	EfficientNetV2-S	61.59± 4.36	40.67± 3.38	20.92	31.44± 3.86	30.15
50-shot	FFN	33.31± 1.01	14.15± 0.71	19.16	9.94± 0.98	23.37
	VGG-16	38.95± 0.61	17.47± 0.96	21.48	10.36± 0.45	28.59
	ResNet-18	39.18± 1.19	17.78± 0.62	21.40	10.64± 0.64	28.54
	DenseNet-121	43.64± 0.68	18.56± 0.43	25.08	10.15± 0.79	33.49
	ViT-B/16	87.92± 0.45	82.77± 1.76	5.15	81.05± 2.08	6.87
	EfficientNetV2-S	74.75± 1.05	59.92± 3.84	14.83	56.17± 3.56	18.58
200-shot	FFN	41.98± 0.79	21.05± 0.28	20.93	13.11± 0.73	28.87
	VGG-16	52.87± 0.78	25.29± 0.46	27.58	15.35± 0.91	37.52
	ResNet-18	53.67± 0.95	25.58± 0.42	28.09	15.87± 0.51	37.80
	DenseNet-121	61.69± 0.36	33.06± 1.57	28.63	19.48± 0.86	42.21
	ViT-B/16	95.30± 0.14	95.77± 0.19	-0.47	95.22± 0.26	0.08
	EfficientNetV2-S	88.25± 0.23	83.61± 1.24	4.64	82.28± 1.95	5.97
2000-shot	FFN	58.83± 1.44	46.19± 0.66	12.64	44.56± 1.86	14.27
	VGG-16	78.50± 0.59	77.58± 0.40	0.92	76.34± 0.55	2.16
	ResNet-18	79.40± 0.35	79.00± 0.37	0.40	78.14± 0.17	1.26
	DenseNet-121	84.65± 0.34	84.70± 0.17	-0.05	83.58± 0.30	1.07
	ViT-B/16	97.86± 0.09	98.06± 0.12	-0.20	98.01± 0.08	-0.15
	EfficientNetV2-S	95.30± 0.18	95.79± 0.09	-0.49	95.07± 0.16	0.23

Table 12: Results on CIFAR100.

shots	Models	Random-Bench Accuracy	Achilles-Bench (GradNorm) Accuracy	Gap	Achilles-Bench (Loss) Accuracy	Gap
20-shot	FFN	10.48 $\pm$ 0.32	5.69 $\pm$ 0.11	4.79	1.90 $\pm$ 0.20	8.58
	VGG-16	18.87 $\pm$ 0.46	10.42 $\pm$ 0.17	8.45	2.77 $\pm$ 0.16	16.10
	ResNet-18	17.20 $\pm$ 0.57	9.01 $\pm$ 0.34	8.19	2.61 $\pm$ 0.11	14.59
	DenseNet-121	21.48 $\pm$ 0.80	10.84 $\pm$ 0.88	10.64	3.24 $\pm$ 0.23	18.24
	ViT-B/16	68.23 $\pm$ 1.68	61.45 $\pm$ 4.99	6.78	54.99 $\pm$ 2.04	13.24
	EfficientNetV2-S	55.10 $\pm$ 0.18	51.87 $\pm$ 1.43	3.23	40.68 $\pm$ 1.30	14.42
200-shot	FFN	23.67 $\pm$ 1.00	16.83 $\pm$ 0.46	6.84	12.91 $\pm$ 1.45	10.76
	VGG-16	45.52 $\pm$ 0.45	41.41 $\pm$ 0.77	4.11	36.22 $\pm$ 0.34	9.30
	ResNet-18	44.37 $\pm$ 0.70	41.03 $\pm$ 1.03	3.34	36.95 $\pm$ 1.02	7.42
	DenseNet-121	53.75 $\pm$ 0.39	51.80 $\pm$ 0.61	1.95	48.04 $\pm$ 0.37	5.71
	ViT-B/16	88.89 $\pm$ 0.24	89.00 $\pm$ 0.21	-0.11	88.86 $\pm$ 0.40	0.03
	EfficientNetV2-S	79.63 $\pm$ 0.64	80.36 $\pm$ 0.45	-0.73	78.32 $\pm$ 0.43	1.31

Table 13: Results on ImageNet.

shots	Models	Random-Bench Accuracy	Achilles-Bench (GradNorm) Accuracy	Gap	Achilles-Bench (Loss) Accuracy	Gap
50-shot	FFN	3.93 $\pm$ 0.51	1.59 $\pm$ 0.09	2.34	1.72 $\pm$ 0.04	2.21
	VGG-16	6.94 $\pm$ 0.43	2.11 $\pm$ 0.23	4.83	2.97 $\pm$ 0.23	3.97
	ResNet-18	18.84 $\pm$ 0.46	11.67 $\pm$ 0.27	7.17	9.46 $\pm$ 0.24	9.38
	DenseNet-121	22.96 $\pm$ 0.44	13.89 $\pm$ 0.45	9.07	10.29 $\pm$ 0.20	12.67

Table 14: The results on GLUE with 16-shots.

Datasets	Models	Random-Bench Accuracy	Achilles-Bench (GradNorm) Accuracy	Gap	Achilles-Bench (Loss) Accuracy	Gap
SST2	Transformer	52.64 $\pm$ 2.16	52.89 $\pm$ 0.56	-0.25	52.34 $\pm$ 0.26	0.30
	BERT	68.39 $\pm$ 7.14	56.86 $\pm$ 4.88	11.53	50.28 $\pm$ 0.88	18.11
	GPT-2	55.62 $\pm$ 4.12	52.52 $\pm$ 2.00	3.10	51.54 $\pm$ 1.22	4.08
	RoBERTa	76.67 $\pm$ 3.44	58.12 $\pm$ 1.47	18.55	50.25 $\pm$ 0.96	26.42
	T5	55.94 $\pm$ 3.74	51.95 $\pm$ 1.90	3.99	51.19 $\pm$ 2.26	4.75
COLA	Transformer	68.95 $\pm$ 0.44	68.74 $\pm$ 0.86	0.21	68.88 $\pm$ 0.50	0.07
	BERT	66.94 $\pm$ 3.55	64.99 $\pm$ 6.39	1.95	58.16 $\pm$ 13.44	8.78
	GPT-2	66.40 $\pm$ 5.50	66.19 $\pm$ 5.92	0.21	66.56 $\pm$ 5.19	-0.16
	RoBERTa	69.66 $\pm$ 1.02	65.23 $\pm$ 5.14	4.43	49.38 $\pm$ 11.73	20.28
	T5	55.82 $\pm$ 8.92	59.54 $\pm$ 4.84	-3.72	56.80 $\pm$ 6.79	-0.98
MNLI	Transformer	35.40 $\pm$ 0.09	35.45 $\pm$ 0.00	-0.05	35.28 $\pm$ 0.21	0.12
	BERT	36.21 $\pm$ 0.96	34.21 $\pm$ 0.54	2.00	34.03 $\pm$ 0.96	2.18
	GPT-2	37.63 $\pm$ 1.29	34.40 $\pm$ 1.39	3.23	33.88 $\pm$ 1.31	3.75
	RoBERTa	43.13 $\pm$ 2.07	35.48 $\pm$ 0.82	7.65	33.38 $\pm$ 1.10	9.75
	T5	33.98 $\pm$ 0.50	33.44 $\pm$ 0.21	0.54	33.39 $\pm$ 0.18	0.59
QNLI	Transformer	53.48 $\pm$ 2.46	50.95 $\pm$ 0.55	2.53	51.22 $\pm$ 0.38	2.26
	BERT	53.75 $\pm$ 0.69	50.79 $\pm$ 0.31	2.96	50.10 $\pm$ 0.66	3.65
	GPT-2	55.16 $\pm$ 3.26	53.65 $\pm$ 2.94	1.51	52.49 $\pm$ 2.08	2.67
	RoBERTa	63.52 $\pm$ 3.92	50.80 $\pm$ 0.37	12.72	49.78 $\pm$ 0.40	13.74
	T5	54.03 $\pm$ 2.36	50.85 $\pm$ 1.02	3.18	49.69 $\pm$ 0.94	4.34
MRPC	Transformer	68.63 $\pm$ 0.31	68.38 $\pm$ 0.00	0.25	68.33 $\pm$ 0.10	0.30
	BERT	66.47 $\pm$ 3.22	63.19 $\pm$ 6.52	3.28	54.61 $\pm$ 16.68	11.86
	GPT-2	67.75 $\pm$ 1.53	66.23 $\pm$ 4.44	1.52	63.87 $\pm$ 8.90	3.88
	RoBERTa	69.26 $\pm$ 1.48	57.60 $\pm$ 7.79	11.66	33.33 $\pm$ 0.83	35.93
	T5	58.58 $\pm$ 5.94	59.90 $\pm$ 4.06	-1.32	56.32 $\pm$ 8.15	2.26
QQP	Transformer	63.75 $\pm$ 0.55	63.23 $\pm$ 0.08	0.52	63.19 $\pm$ 0.02	0.56
	BERT	64.81 $\pm$ 2.15	59.19 $\pm$ 2.86	5.62	57.27 $\pm$ 4.69	7.54
	GPT-2	62.57 $\pm$ 1.34	54.64 $\pm$ 4.89	7.93	56.84 $\pm$ 3.40	5.73
	RoBERTa	65.55 $\pm$ 1.36	63.18 $\pm$ 0.00	2.37	63.10 $\pm$ 0.10	2.45
	T5	55.49 $\pm$ 3.35	56.61 $\pm$ 3.44	-1.12	56.14 $\pm$ 2.58	-0.65
RTE	Transformer	53.72 $\pm$ 0.90	54.95 $\pm$ 1.01	-1.23	54.80 $\pm$ 0.42	-1.08
	BERT	55.02 $\pm$ 1.56	53.43 $\pm$ 2.41	1.59	50.40 $\pm$ 2.59	4.62
	GPT-2	58.77 $\pm$ 3.98	58.84 $\pm$ 2.84	-0.07	52.71 $\pm$ 1.69	6.06
	RoBERTa	55.16 $\pm$ 1.73	53.14 $\pm$ 0.42	2.02	52.56 $\pm$ 0.37	2.60
	T5	51.05 $\pm$ 2.44	49.03 $\pm$ 1.76	2.02	49.10 $\pm$ 0.97	1.95
WNLI	Transformer	58.03 $\pm$ 2.07	56.62 $\pm$ 0.56	1.41	57.46 $\pm$ 1.05	0.57
	BERT	56.34 $\pm$ 6.96	54.37 $\pm$ 4.51	1.97	56.62 $\pm$ 2.87	-0.28
	GPT-2	56.34 $\pm$ 0.00	56.62 $\pm$ 1.38	-0.28	56.90 $\pm$ 2.61	-0.56
	RoBERTa	57.75 $\pm$ 3.09	56.90 $\pm$ 1.13	0.85	56.34 $\pm$ 1.54	1.41
	T5	58.31 $\pm$ 0.69	53.52 $\pm$ 5.12	4.79	52.11 $\pm$ 5.42	6.20

Table 15: Results on GLUE with 32-shots.

Datasets	Models	Random-Bench Accuracy	Achilles-Bench (GradNorm) Accuracy	Gap	Achilles-Bench (Loss) Accuracy	Gap
SST2	Transformer	55.96 $\pm$ 1.33	52.50 $\pm$ 0.58	3.46	52.27 $\pm$ 0.48	3.69
	BERT	77.20 $\pm$ 4.97	54.70 $\pm$ 4.07	22.50	50.28 $\pm$ 0.85	26.92
	GPT-2	68.46 $\pm$ 5.20	57.73 $\pm$ 1.01	10.73	51.72 $\pm$ 0.72	16.74
	RoBERTa	83.81 $\pm$ 2.25	57.36 $\pm$ 2.45	26.45	50.09 $\pm$ 0.97	33.72
	T5	63.10 $\pm$ 3.97	54.06 $\pm$ 2.86	9.04	51.06 $\pm$ 2.38	12.04
COLA	Transformer	69.36 $\pm$ 0.37	69.13 $\pm$ 0.00	0.23	69.15 $\pm$ 0.04	0.21
	BERT	67.56 $\pm$ 3.19	62.05 $\pm$ 9.21	5.51	60.44 $\pm$ 10.81	7.12
	GPT-2	66.94 $\pm$ 3.91	66.06 $\pm$ 5.77	0.88	66.04 $\pm$ 6.22	0.90
	RoBERTa	72.23 $\pm$ 1.32	66.27 $\pm$ 3.90	5.96	59.85 $\pm$ 12.56	12.38
	T5	59.50 $\pm$ 4.25	58.39 $\pm$ 5.32	1.11	57.81 $\pm$ 6.12	1.69
MNLI	Transformer	35.46 $\pm$ 0.03	35.45 $\pm$ 0.00	0.01	35.42 $\pm$ 0.04	0.04
	BERT	39.21 $\pm$ 2.50	34.45 $\pm$ 0.76	4.76	33.41 $\pm$ 0.63	5.80
	GPT-2	39.81 $\pm$ 0.88	34.75 $\pm$ 1.47	5.06	34.00 $\pm$ 1.28	5.81
	RoBERTa	45.44 $\pm$ 2.02	36.55 $\pm$ 0.97	8.89	33.81 $\pm$ 1.18	11.63
	T5	34.45 $\pm$ 0.50	33.87 $\pm$ 0.24	0.58	33.13 $\pm$ 0.24	1.32
QNLI	Transformer	53.96 $\pm$ 0.92	51.36 $\pm$ 0.40	2.60	50.78 $\pm$ 0.09	3.18
	BERT	57.07 $\pm$ 2.02	50.85 $\pm$ 0.40	6.22	50.08 $\pm$ 0.35	6.99
	GPT-2	57.97 $\pm$ 2.83	53.70 $\pm$ 3.03	4.27	52.86 $\pm$ 2.50	5.11
	RoBERTa	71.64 $\pm$ 1.99	50.67 $\pm$ 0.62	20.97	49.61 $\pm$ 0.31	22.03
	T5	60.41 $\pm$ 4.20	50.74 $\pm$ 1.29	9.67	49.50 $\pm$ 1.04	10.91
MRPC	Transformer	68.63 $\pm$ 0.27	68.43 $\pm$ 0.10	0.20	68.48 $\pm$ 0.20	0.15
	BERT	66.96 $\pm$ 2.36	61.72 $\pm$ 7.05	5.24	54.80 $\pm$ 16.04	12.16
	GPT-2	69.07 $\pm$ 1.82	67.11 $\pm$ 2.33	1.96	63.97 $\pm$ 8.95	5.10
	RoBERTa	73.73 $\pm$ 2.79	55.34 $\pm$ 6.31	18.39	35.39 $\pm$ 4.57	38.34
	T5	62.21 $\pm$ 3.44	58.04 $\pm$ 5.53	4.17	55.78 $\pm$ 8.91	6.43
QQP	Transformer	64.06 $\pm$ 0.30	63.26 $\pm$ 0.09	0.80	63.18 $\pm$ 0.00	0.88
	BERT	65.49 $\pm$ 1.73	61.12 $\pm$ 1.72	4.37	54.83 $\pm$ 5.47	10.66
	GPT-2	63.37 $\pm$ 3.21	56.03 $\pm$ 4.17	7.34	55.00 $\pm$ 5.30	8.37
	RoBERTa	70.10 $\pm$ 0.98	61.82 $\pm$ 2.73	8.28	62.81 $\pm$ 0.45	7.29
	T5	61.44 $\pm$ 4.99	56.21 $\pm$ 2.96	5.23	54.53 $\pm$ 4.05	6.91
RTE	Transformer	53.72 $\pm$ 0.98	55.38 $\pm$ 0.49	-1.66	55.02 $\pm$ 0.74	-1.30
	BERT	55.02 $\pm$ 3.83	52.85 $\pm$ 1.96	2.17	49.46 $\pm$ 2.45	5.56
	GPT-2	58.77 $\pm$ 2.65	60.36 $\pm$ 2.96	-1.59	52.85 $\pm$ 3.15	5.92
	RoBERTa	57.76 $\pm$ 3.62	54.95 $\pm$ 2.37	2.81	52.78 $\pm$ 0.58	4.98
	T5	51.48 $\pm$ 1.13	52.06 $\pm$ 1.77	-0.58	49.17 $\pm$ 1.60	2.31
WNLI	Transformer	58.59 $\pm$ 2.76	56.34 $\pm$ 0.00	2.25	58.31 $\pm$ 1.44	0.28
	BERT	54.08 $\pm$ 3.94	54.93 $\pm$ 4.27	-0.85	55.21 $\pm$ 2.87	-1.13
	GPT-2	58.03 $\pm$ 2.25	57.46 $\pm$ 1.87	0.57	56.34 $\pm$ 1.99	1.69
	RoBERTa	56.62 $\pm$ 0.56	56.90 $\pm$ 1.13	-0.28	57.18 $\pm$ 1.44	-0.56
	T5	53.80 $\pm$ 3.92	57.18 $\pm$ 3.03	-3.38	53.24 $\pm$ 5.52	0.56

Table 16: Results on GLUE with 100-shots.

Datasets	Models	Random-Bench Accuracy	Achilles-Bench (GradNorm) Accuracy	Gap	Achilles-Bench (Loss) Accuracy	Gap
SST2	Transformer	59.50± 1.52	52.41± 0.64	7.09	51.74± 0.38	7.76
	BERT	86.22± 0.39	51.38± 1.86	34.84	49.33± 2.12	36.89
	GPT-2	83.00± 1.43	53.46± 1.76	29.54	51.22± 1.85	31.78
	RoBERTa	88.37± 1.08	51.93± 0.73	36.44	50.57± 0.75	37.80
	T5	83.35± 4.21	52.25± 1.54	31.10	51.01± 2.49	32.34
COLA	Transformer	69.19± 0.08	69.17± 0.05	0.02	68.78± 0.69	0.41
	BERT	74.84± 1.36	61.25± 7.11	13.59	57.09± 12.93	17.75
	GPT-2	66.62± 3.15	65.77± 5.94	0.85	64.60± 6.87	2.02
	RoBERTa	77.28± 1.09	62.84± 6.71	14.44	59.64± 5.95	17.64
	T5	75.24± 1.07	57.49± 5.44	17.75	56.72± 7.18	18.52
MNLI	Transformer	35.61± 0.22	35.25± 0.32	0.36	35.11± 0.41	0.50
	BERT	43.98± 2.71	34.74± 0.62	9.24	33.13± 0.60	10.85
	GPT-2	48.69± 1.80	34.77± 1.12	13.92	33.86± 1.24	14.83
	RoBERTa	61.44± 2.69	36.87± 1.00	24.57	33.64± 0.98	27.80
	T5	40.63± 4.32	34.27± 0.36	6.36	32.98± 0.29	7.65
QNLI	Transformer	56.23± 0.86	50.68± 0.15	5.55	50.54± 0.00	5.69
	BERT	63.82± 4.63	49.98± 1.01	13.84	47.26± 2.17	16.56
	GPT-2	62.52± 4.58	53.34± 2.93	9.18	51.80± 2.02	10.72
	RoBERTa	78.44± 2.05	50.08± 0.51	28.36	50.23± 0.46	28.21
	T5	73.75± 2.43	49.99± 1.01	23.76	48.91± 1.36	24.84
MRPC	Transformer	69.02± 0.69	68.43± 0.10	0.59	66.67± 3.43	2.35
	BERT	69.41± 0.95	58.77± 5.61	10.64	48.24± 11.18	21.17
	GPT-2	71.76± 1.91	65.34± 2.14	6.42	64.07± 5.23	7.69
	RoBERTa	77.16± 2.77	60.34± 4.15	16.82	41.47± 6.42	35.69
	T5	65.64± 1.33	57.79± 5.58	7.85	56.52± 5.77	9.12
QQP	Transformer	65.27± 0.61	63.45± 0.32	1.82	60.54± 1.67	4.73
	BERT	69.62± 1.85	60.11± 1.68	9.51	47.14± 5.08	22.48
	GPT-2	70.13± 2.28	55.04± 4.57	15.09	51.27± 6.85	18.86
	RoBERTa	75.33± 1.24	63.02± 1.13	12.31	48.10± 12.34	27.23
	T5	73.02± 1.75	57.54± 4.38	15.48	53.35± 4.40	19.67
RTE	Transformer	53.72± 0.90	56.10± 0.71	-2.38	53.07± 0.23	0.65
	BERT	54.66± 2.65	52.56± 1.06	2.10	47.44± 1.49	7.22
	GPT-2	59.35± 3.23	57.76± 2.33	1.59	51.26± 2.40	8.09
	RoBERTa	63.39± 2.49	53.94± 0.67	9.45	51.05± 2.19	12.34
	T5	53.72± 3.97	51.19± 1.59	2.53	48.59± 1.32	5.13

G Visualization

G.1 Selected Sentences by Achilles-Bench for NLP tasks

Table 17: Sentences of the set random selected on SST2. We sample randomly 10 examples for each label.

sentence	label
inconsistent , meandering , and sometimes dry plot made a great saturday night live sketch , but a great movie it is not an mtv , sugar hysteria , was only it 's been 13 months and 295 preview screenings since i last walked out on a movie , but resident evil really earned my indignant , preemptive departure act weird humbuggery ... 90 punitive minutes of eardrum-dicing gunplay , screeching-metal smashups , and flaccid odd-couple sniping . of screenwriting cliches that sink it faster than a leaky freighter sit still for two hours and change watching such a character , especially when rendered in as flat and impassive a manner as phoenix 's	negative
a smart , solid , kinetically-charged spy flick worthy of a couple hours of summertime and a bucket of popcorn great acting have ever seen , constantly pulling the rug from underneath us , seeing things from new sides , plunging deeper , getting more intense is a film in which the talent is undeniable come away with a greater knowledge of the facts of cuban music shows how deeply felt emotions can draw people together across the walls that might otherwise separate them . the crazy things that keep people going in this crazy life appeal to asian cult cinema fans and asiaphiles interested to see what all the fuss is about . potentially interesting thrusts the audience	positive

Table 18: Sentences of the set searched by Achilles-Bench (GradNorm) for SST2. We choose top 10 examples for each label.

sentence	label
is well below expectations . make it sting is well below expectations huge sacrifice best spent elsewhere few ' cool ' actors laughably below is well below expectations . spare dialogue temperamental	negative
to winger fans who have missed her since 1995 's forget paris rocky and becomes compulsively watchable particularly balk , who 's finally been given a part worthy of her considerable talents balk , who 's finally been given a part worthy of her considerable talents clearly a manipulative film entertainingly nasty busts out of its comfy little cell fascinate me rediscovers his passion in life	positive

Table 19: Sentences of the set searched by Achilles-Bench (Loss) for SST2. We choose top 10 examples for each label.

sentence	label
a damn fine and a truly distinctive and a deeply pertinent film provides an invaluable service is an undeniably worthy and devastating experience gain the unconditional love she seeks unfolds as one of the most politically audacious films of recent decades from any country , but especially from france self-deprecating , biting and witty feature reasonably creative eighth-grader chilling tale noble end from sharing the awe in which it holds itself	negative
fails to have a heart , mind or humor of its own terminally bland , 's not a brilliant piece of filmmaking after next spreads them pretty thin an admittedly middling film the movie is silly beyond comprehension , just a bunch of good actors flailing around in a caper that 's neither original nor terribly funny , incoherence and sub-sophomoric he script is n't up to the level of the direction an overcooked souffl	positive

G.2 Visualization of the “Average Examples (Random)”

Figure 8: Visualization of the set random selected on CIFAR-10. We sample randomly 50 examples for each label.



G.3 Visualization of the Searched “Hard Examples (GradNorm)”

Figure 9: Visualization of the set searched by Achilles-Bench (GradNorm) on CIFAR-10. We choose top 50 examples for each label.



G.4 Visualization of the Searched “Hard Examples (Loss)”

Figure 10: Visualization of the set searched by Achilles-Bench (Loss) on CIFAR-10. We choose top 50 examples for each label.

